

Algorithm Complexity

If $g(n) = O(f(n))$, then $c g(n) = O(f(n))$ for any constant c .

If $g_1(n)$ and $g_2(n)$ are both $O(f(n))$, then so is $g_1(n) + g_2(n)$.

If $g_1(n) = O(f(n))$ but $g_2(n) = o(f(n))$, then $g_1(n) + g_2(n) = O(f(n))$.

If $g(n) = O(f(n))$, and $f(n) = o(h(n))$, then $g(n) = o(h(n))$.

Number Theory

Division algorithm: Let x be an integer and let N be a positive integer. Then, there are unique integers q and r , with $0 \leq r \leq N - 1$, such that $x = qN + r$. The number q is the quotient and the r is called the remainder: $q = N \text{ div } d$; $r = N \text{ mod } d$. If the remainders of x and y are the same when divided by d , then $x \pmod{N} \equiv y \pmod{N}$ if $N | (x - y)$.

Congruence identities: $x \cdot y \pmod{N} \equiv x \pmod{N} \cdot y \pmod{N}$; $x + y \pmod{N} \equiv x \pmod{N} + y \pmod{N}$.

Multiplicative inverse: A multiplicative inverse $\text{mod } N$ of an integer x is an integer $s \in \{1, 2, \dots, n - 1\}$ such that $sx \pmod{N} = 1$.

RSA key generation: Given two prime numbers p and q , set $N = p \cdot q$ and $\phi = (p - 1) \cdot (q - 1)$. Then e is a number chosen to be coprime to ϕ and d is the inverse of $e \pmod{\phi}$. The public information is N and e , and the private information is p, q, ϕ, d . Encryption: $c = m^e \pmod{N}$; decryption: $m = c^d \pmod{N}$.

Recursion and Induction

Arithmetic sequence: $\sum_{k=0}^{n-1} (a + kd) = an + \frac{d(n-1)n}{2}$

Geometric sequence: $\sum_{k=0}^{n-1} a \cdot r^k = \frac{a(r^n - 1)}{r - 1}$; $\sum_{k=0}^{\infty} a \cdot r^k = \frac{a}{r - 1}$

The characteristic equation of a linear homogeneous recurrence relation $f_n = c_1 f_{n-1} + c_2 f_{n-2} + \dots + c_k f_{n-k}$ is given by $x^n = c_1 x^{n-1} + c_2 x^{n-2} + \dots + c_k x^{n-k}$. If the roots of the characteristic equation are $r_1, \dots, r_k$ without multiplicity, then the general solution to the linear homogeneous recurrence relation is $f_n = m_1 \cdot r_1^n + m_2 \cdot r_2^n + \dots + m_k \cdot r_k^n$.

Selected Algorithms

Euclidean algorithm

```
function gcd(a, b)
  while b ≠ 0
    t := b;
    b := a mod b;
    a := t;
  return a;
```

Fast exponentiation algorithm

```
p := 1
s := x
r := y

While ( r > 0 )
  If ( r mod 2 = 1 )
    p := p · s
  s := s · s
  r := r div 2
End-while

Return (p)
```

Advanced Counting

Combinations: ${}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$

Permutations: ${}_nP_k = \frac{n!}{(n-k)!}$

Binomial theorem: $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$.

Multiset counting: The number of ways to select n objects from a set of m varieties is $\binom{n+m-1}{m-1}$.

Discrete Probability

Expected Value The *expected value* of a discrete random variable X is $E[X] = \sum_{s \in S} X(s)p(s)$ where S is the set of possible outcomes of X , $p(s)$ is the probability of outcome s , and $X(s)$ is the value of X for s .

Conditional probability: $P(F|X) = \frac{P(X \cap F)}{P(X)}$ **Bayes' Theorem:** $P(F|X) = \frac{P(X|F) \cdot P(F)}{P(X|F) \cdot P(F) + P(X|\bar{F}) \cdot P(\bar{F})}$